Data Description

Data Sources & Final Dataset

Our final analysis dataset is built from **three** integrated sources. (1) Spotify Global Weekly Top 50 charts provide rank and stream-level popularity over five years (January 7, 2021 to February 19, 2026). (2) We extract nine acoustic features per unique track using Librosa on YouTube-sourced audio snippets. (3) Following interview feedback (added after Phase 2), we enrich each track with artist-level popularity metrics, featured-artist counts and artists' record labels. All three are merged on track URI / artist names to produce one row per (track, week) chart appearance.

Dataset 1 — Spotify Global Weekly Charts

Notebook: link

- Rows: one track's appearance on a given week's Global Top 50.
- Columns we focus on: *uri* (merge key), *track_name*, *artist_names*, *streams*, *peak_rank*, *weeks_on_chart*, and *date* for longitudinal analysis.
- Why relevant: provides objective, market-level success metrics (streams, weeks_on_chart, peak_rank) that operationalize song longevity for both research questions.
- Source: compiled and published by Spotify from real-time streaming activity.
- Acquisition: Selenium-based scraper handles authentication and downloads weekly CSVs; files are batched by month and concatenated.

Dataset 2 — Librosa Audio Features

Notebook : link

- Rows: one row per unique track (deduplicated by URI before feature extraction).
- Columns we focus on: tempo, energy, zero_crossing_rate, spectral_centroid, spectral_rolloff, mfcc_1, mfcc_2, chroma_mean, chroma_std.
- Why relevant: translates the subjective sound of a song into nine quantitative descriptors so we can compare acoustic profiles across hits and across time (RQ1, RQ2).
- Acquisition: yt-dlp retrieves the first 60 seconds of audio per track from YouTube; Librosa computes feature means.
- Caveat: features depend on whichever YouTube upload is matched first, which can introduce noise from intros, dialogue, or non-studio versions.

Dataset 3 — Artist-Level Enrichment (NEW, post-Phase 2)

Notebook : link

Based on interview feedback, we incorporated additional artist-level data to better contextualize a song's success beyond its acoustic profile.

- artist_listener_count — monthly listeners as a proxy for an artist's overall reach.
- artist_popularity_score — Spotify's 0–100 popularity index, used as a second proxy for artist-level fame.
- num_featured_artists — count of featured artists on each track, letting us distinguish solo songs from collaborations.
- Why relevant: these features control for artist-driven popularity, isolating the contribution of acoustic content to chart success and addressing a key concern raised in user interviews.

Dataset 4 — Record Label Enrichment (NEW, post-Phase 2)
Notebook: link (5.2 H2 — Label power & longevity cell)
Additional dataset only for hypothesis 2

To investigate whether major label affiliation influences chart longevity, we first needed a reliable way to classify each track's record label. We adopted the industry-standard Big Three framework, including Universal Music Group (UMG), Sony Music Entertainment (SME), and Warner Music Group (WMG), as the benchmark for major label status.

This classification is well-supported by market data. According to Music & Copyright's 2024 annual survey, the Big Three collectively account for approximately 69.5% of global recorded music revenue (digital and physical combined): UMG leads with 31.7%, followed by SME at 22.5% and WMG at 15.3%, with independent labels accounting for the remaining 30.5%.

Beyond market share, these three companies have the infrastructure, such as marketing budgets, distribution networks, and playlist relationships, that independent labels simply can't match, which is exactly the kind of structural advantage we're trying to capture in our hypothesis.

- **Label Classification Logic**
  The source field in the dataset represents the record label that released each track. To analyze the effect of major label affiliation, each source was classified into one of three categories:
  - **0 – Major Label (Parent Company):** The release is directly attributed to one of the three major music conglomerates themselves: Sony Music Entertainment, Universal Music Group, or Warner Music Group and their regional divisions (e.g., Sony Music Latin, Universal Music LLC, Warner Music India).
  - **1 – Major Label Subsidiary:** The release comes from a sub-label or imprint owned by one of the three majors. For example, Columbia Records is a Sony subsidiary, Republic Records belongs to Universal Music Group, and Atlantic Records is under Warner Music Group.
  - **2 – Independent / Other:** The release does not belong to any of the three majors or their known subsidiaries.
- **Classification Process**
  Rather than manually classifying all 13,399 rows, the process was optimized as follows:
  - The source field was deduplicated to extract **350 unique label names**.
  - An LLM (Claude) was used to classify each unique source into label 0, 1, or 2 based on known industry knowledge of major label ownership structures, producing a lookup table (raw_unique_source_labeled.csv).
  - This labeled lookup table was then **merged back** into the full dataset on the source field, assigning a major_label value to all 13,399 rows efficiently.
- **Resulting Distribution (row-level)**

Label	Description	Count

0	Major label parent companies	832
1	Major label subsidiaries	8,015
2	Independent / other	4,552

- This approach ensures consistency across all rows while minimizing manual effort, since multiple rows can share the same source, classifying at the unique-source level and merging is far more efficient than reviewing each row individually.

All column definitions :

Column Name	**Description**
rank	Current position of the track on the chart for the given date
uri	Unique identifier for the track (often Spotify URI)
artist_names	Name(s) of the artist(s) performing the track
track_name	Title of the song/track
source	Platform or source of the chart data (e.g., Spotify, Billboard, etc.)
peak_rank	Highest rank the track has achieved historically on the chart
previous_rank	The track's rank in the previous time period (e.g., previous day/week)
weeks_on_chart	Total number of weeks the track has appeared on the chart
streams	Number of streams the track received in the given period
date	The specific date of the chart snapshot
tempo	Estimated BPM (beats per minute)
energy	Mean RMS energy (loudness proxy)
zero_crossing_rate	Rate of sign changes in the audio signal (noisiness proxy)
spectral_centroid	Spectral "brightness" or perceived pitch center
spectral_rolloff	Frequency below which 85% of signal energy is contained
mfcc_1, mfcc_2	First two Mel-frequency cepstral coefficients representing timbral features
chroma_mean	Average chroma energy (harmonic content distribution)
chroma_std	Standard deviation of chroma energy (harmonic variation)

monthly_listener	Number of monthly listeners of the primary artist (sourced from kworb.net Spotify listeners data)
artist_count	Number of artists featured on the track
collaboration_type	Binary indicator: 0 = solo artist, 1 = collaboration (multiple artists)